

## Module 2

- While loop:-

A while loop is an entry-controlled loop. It checks the condition before executing the loop body. If the condition is true, the loop runs otherwise, it stops.

Syntax →

```
while (condition)
{
    // Statement
}
```

Uses of while loop →

- Counting digits
- Reversing numbers
- Sum of digits
- factorial
- menu-driven programs.

# for loop

A for loop is an entry-controlled loop used when the number of iterations is known.

Syntax →

~~for (condition)~~

for (initialization; condition; increment/  
decrement)

{  
// Statement  
}

Uses of for loop:-

- Printing pattern
- printing factors
- Prime number checking
- Tables
- Arrays.

**Problem Statement:**

Given a positive integer **N**, determine the total number of digits present in the number.

Your task is to write a program that counts and prints the total number of digits.

**Input Format**

A single integer **N**.

**Output Format**

Print a single integer representing the total number of digits in **N**.

**Constraints**

$$1 \leq N \leq 10^{18}$$

**Example 1****Input**

12345

**Output**

5

**Explanation**

The number **12345** contains **5** digits.

**Example 2****Input**

7

**Output**

1

**Explanation**

The number **7** consists of only one digit.



FILE EXPLORER

week1.cpp week3.cpp week4.cpp

CPP\_MODULES

tempCodeRunner...  
tempCodeRunner...  
week1.cpp  
week3.cpp  
week4.cpp

week4.cpp

```
1 // QUESTION 1  
2 #include <iostream>  
3 using namespace std;  
4  
5 int main() {  
6     int N, count = 0;  
7  
8     cin >> N;  
9  
10    while (N > 0) {  
11        count++;  
12        N = N / 10;  
13    }  
14  
15    cout << count;  
16  
17    return 0;  
18 }  
19  
20
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
cd "/Users/apple/Documents/Cpp_modules/" && g++ tempCodeRunnerFile.cpp -o tempCodeRunnerFile && "/Users/apple/Documents/Cpp_modules/"tempCodeRunnerFile  
apple@apples-MacBook-Pro Cpp_modules % cd "/Users/apple/Documents/Cpp_modules/" && g++ tempCodeRunnerFile.cpp -o tempCodeRunnerFile && "/Users/apple/Documents/Cpp_modules/"t  
empCodeRunnerFile  
123456789  
apple@apples-MacBook-Pro Cpp_modules %
```

Tip: Try the Plan  
agent to research and  
plan before  
implementing  
changes.

week4.cpp:2-19

**Problem Statement**

Given a positive integer **N**, reverse its digits and print the reversed number.

Your task is to write a program that reverses the given number.

**Input Format**

A single integer **N**.

**Output Format**

Print the reversed number.

**Constraints**

$$1 \leq N \leq 10^{18}$$

**Example 1****Input**

12345

**Output**

54321

**Explanation**

Reversing the digits of **12345** gives **54321**.

**Example 2****Input**

90010

**Output**

1009

**Explanation**

After reversing the digits, the leading zeros are removed automatically, resulting in **1009**.



MODULES

C- week4.cpp > main()

```

20
21 //question 2
22 #include <iostream>
23 using namespace std;
24
25 int main() {
26     int N, reverse = 0, digit;
27
28     cin >> N;
29
30     while (N > 0) {
31         digit = N % 10;
32         reverse = reverse * 10 + digit;
33         N = N / 10;
34     }
35
36     cout << reverse;
37
38     return 0;
39 }
40
41

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```

cd "/Users/apple/Documents/CPP_modules/" && g++ tempCodeRunnerFile.cpp -o tempCodeRunnerFile && "/Users/apple/Documents/CPP_modules/"tempCodeRunnerF
apple@apples-MacBook-Pro CPP_modules % cd "/Users/apple/Documents/CPP_modules/" && g++ tempCodeRunnerFile.cpp -o tempCodeRunnerFile && "/Users/apple
tempCodeRunnerFile
3579
9753
apple@apples-MacBook-Pro CPP_modules %

```

**Problem Statement**

Given a positive integer **N**, calculate the sum of all its digits.

Your task is to write a program that computes and prints the sum of the digits.

**Input Format**

A single integer **N**.

**Output Format**

Print a single integer representing the sum of all digits.

**Constraints**

$$1 \leq N \leq 10^{18}$$

**Example 1****Input**

12345

**Output**

15

**Explanation**

The sum of the digits is:

$$1 + 2 + 3 + 4 + 5 = 15$$

**Example 2****Input**

9087

**Output**

24

**Explanation**

The sum of the digits is:

$$9 + 0 + 8 + 7 = 24$$

EXPLORER

week3\_main.c

week4\_main.c

C\_AND\_CPP\_MODULES

tempCodeRunner...

tempCodeRunner...

week2\_main.c

week3\_main.c

week4\_main.c

week4\_main.c > main()

```
43 //question 3
44 #include <stdio.h>
45
46 int main() {
47     int N, digit, sum = 0;
48
49     scanf("%d", &N);
50
51     while (N > 0) {
52         digit = N % 10;
53         sum = sum + digit;
54         N = N / 10;
55     }
56
57     printf("%d", sum);
58
59     return 0;
60 }
61
```

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

```
cd "/Users/apple/Documents/C_AND_CPP_MODULES/" && gcc tempCodeRunnerFile.c -o tempCodeRunnerFile && "/Users/apple/Documents/C_AND_CPP_MODULES/"tempCodeRunnerFile
apple@apples-MacBook-Pro C_AND_CPP_MODULES % cd "/Users/apple/Documents/C_AND_CPP_MODULES/" && gcc tempCodeRunnerFile.c -o tempCodeRunnerFile && "/Users/apple/Documents/C_AND_CPP_MODULES/"tempCodeRunnerFile
4567
222
apple@apples-MacBook-Pro C_AND_CPP_MODULES %
```

Tip: Try the Plan agent to research plan before implementing changes.

C week4\_main

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EXPLORER

week1.cppweek3.cppweek4.cpp

CPP\_MODULES

tempCodeRunner...tempCodeRunner...week1.cppweek3.cppweek4.cpp

41

42 //question 3

43 #include <iostream>

44 using namespace std;

45

46 int main(){

47 int N, digit, sum = 0;

48

49 cin >> N;

50

51 while (N > 0) {

52 digit = N % 10;

53 sum = sum + digit;

54 N = N / 10;

55 }

56

57 cout << sum;

58

59 return 0;

60 }

61

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

Code + -

cd "/Users/apple/Documents/CPP\_modules/" && g++ tempCodeRunnerFile.cpp -o tempCodeRunnerFile && "/Users/apple/Documents/CPP\_modules/"tempCodeRunnerFile

apple@apples-MacBook-Pro CPP\_modules % cd "/Users/apple/Documents/CPP\_modules/" && g++ tempCodeRunnerFile.cpp -o tempCodeRunnerFile && "/Users/apple/Documents/CPP\_modules/"tempCodeRunnerFile

87654

38

apple@apples-MacBook-Pro CPP\_modules %

Tip: Try the agent to reason before implementing changes.

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## Problem Statement

Given a positive integer **N**, count how many **even** digits and how many **odd** digits are present in the number.

Your task is to write a program that prints the total count of even digits followed by the total count of odd digits.

## Input Format

A single integer **N**.

## Output Format

Print two integers separated by a space:

- Number of even digits
- Number of odd digits

## Constraints

$$1 \leq N \leq 10^{18}$$

## Example 1

### Input

123456

### Output

3 3

### Explanation

Even digits are **2, 4, 6**.

Odd digits are **1, 3, 5**.

Therefore, the output is **3 3**.

## Example 2

### Input

908172

### Output

3 3

### Explanation

Even digits are **0, 8, 2**.

Odd digits are **9, 1, 7**.

Therefore, the output is **3 3**.



CPP\_modules

week1.cpp week3.cpp week4.cpp

tempCodeRunner... tempCodeRunner... week1.cpp week3.cpp week4.cpp

46 int main() {

61

62 //question 4

63 #include <iostream>

64 using namespace std;

65

66 int main() {

67 int N, digit;

68 int even = 0, odd = 0;

69

70 cin >> N;

71

72 while (N > 0) {

73 digit = N % 10;

74

75 if (digit % 2 == 0)

76 even++;

77 else

78 odd++;

79

80 N = N / 10;

81 }

82

83 cout << even << " " << odd;

84

85 return 0;

86 }

87

88

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

cd "/Users/apple/Documents/CPP\_modules/" && g++ tempCodeRunnerFile.cpp -o tempCodeRunnerFile && "/Users/apple/Documents/CPP\_modules/"tempCodeRunnerFile

apple@apples-MacBook-Pro CPP\_modules % cd "/Users/apple/Documents/CPP\_modules/" && g++ tempCodeRunnerFile.cpp -o tempCodeRunnerFile && "/Users/apple/Documents/CPP\_modules/"tempCodeRunnerFile

257486

4 2

apple@apples-MacBook-Pro CPP\_modules %

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week4.c

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+ @ ...

### **Problem Statement**

Given a positive integer **N**, find and print the largest digit present in the number.

Your task is to write a program that determines the maximum digit.

### **Input Format**

A single integer **N**.

### **Output Format**

Print the largest digit present in the given number.

### **Constraints**

$$1 \leq N \leq 10^{18}$$

### **Example 1**

#### **Input**

472953

#### **Output**

9

#### **Explanation**

The digits are **4, 7, 2, 9, 5, 3**.

The largest digit among them is **9**.

### **Example 2**

#### **Input**

100248

#### **Output**

8

#### **Explanation**

The digits are **1, 0, 0, 2, 4, 8**.

The largest digit is **8**.

EXPLORER

C AND CPP MODULES

tempCodeRunner...  
tempCodeRunner...  
week2\_main.c  
week3\_main.c  
week4\_main.c

week3\_main.c

week4\_main.c X

week4\_main.c > main()

```
91
92
93 //question 5
94 #include <stdio.h>
95 *
96 int main() {
97     int N, digit;
98     int max = 0;
99
100     scanf("%d", &N);
101
102     while (N > 0) {
103         digit = N % 10;
104
105         if (digit > max)
106             max = digit;
107
108         N = N / 10;
109     }
110
111     printf("%d", max);
112
113     return 0;
114 }
115
116
```

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

Code + -

cd "/Users/apple/Documents/C\_AND\_CPP\_MODULES/" && gcc tempCodeRunnerFile.c -o tempCodeRunnerFile && "/Users/apple/Documents/C\_AND\_CPP\_MODULES/"tempCodeRunnerFile

apple@apples-MacBook-Pro C\_AND\_CPP\_MODULES % cd "/Users/apple/Documents/C\_AND\_CPP\_MODULES/" && gcc tempCodeRunnerFile.c -o tempCodeRunnerFile && "/Users/apple/Documents/C\_AND\_CPP\_MODULES/"tempCodeRunnerFile

85134

8

apple@apples-MacBook-Pro C\_AND\_CPP\_MODULES %

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week4\_main.c

Describe what to

EXPLORER

week1.cpp week3.cpp week4.cpp

CPP\_MODULES

tempCodeRunner... tempCodeRunner... week1.cpp week3.cpp week4.cpp

week4.cpp > main()

```
89 //question 5
90 #include <iostream>
91 using namespace std;
92
93 int main(){
94     int N, digit;
95     int max = 0;
96
97     cin >> N;
98
99     while (N > 0){
100         digit = N % 10;
101
102         if (digit > max)
103             max = digit;
104
105         N = N / 10;
106     }
107
108     cout << max;
109
110     return 0;
111 }
112
```

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

```
cd "/Users/apple/Documents/CPP_modules/" && g++ tempCodeRunnerFile.cpp -o tempCodeRunnerFile && "/Users/apple/Documents/CPP_modules/"tempCodeRunnerFile
apple@apples-MacBook-Pro CPP_modules % cd "/Users/apple/Documents/CPP_modules/" && g++ tempCodeRunnerFile.cpp -o tempCodeRunnerFile && "/Users/apple/Documents/CPP_modules/"tempCodeRunnerFile
754321
apple@apples-MacBook-Pro CPP_modules %
```

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week Describe +

## MEDIUM LEVEL:

### Problem Statement

Given two positive integers **L** and **R**, count how many prime numbers are present in the inclusive range **[L, R]**.

A **prime number** is a natural number greater than **1** that has exactly two distinct positive divisors: **1** and itself.

Your task is to write a program that counts and prints the total number of prime numbers in the given range.

### Input Format

Two integers **L** and **R** separated by a space.

### Output Format

Print a single integer representing the total number of prime numbers in the given range.

### Constraints

$$1 \leq L \leq R \leq 10^5$$

### Example 1

#### Input

1 10

#### Output

4

#### Explanation

The prime numbers between **1** and **10** are: 2, 3, 5, 7

Hence, the total count is **4**.

### Example 2

#### Input

11 20

#### Output

4

#### Explanation

The prime numbers between **11** and **20** are: 11, 13, 17, 19

Hence, the total count is **4**.

EXPLORER

C\_AND\_CPP\_MODULES

tempCodeRunner...  
tempCodeRunner...  
week2\_main.c  
week3\_main.c  
week4\_main.c

week3\_main.c

week4\_main.c X

week4\_main.c > main()

```
118 //question 6
119 #include <stdio.h>
120
121 int main() {
122     int L, R, i, j, count = 0, prime;
123
124     scanf("%d %d", &L, &R);
125
126     for (i = L; i <= R; i++) {
127         if (i < 2)
128             continue;
129
130         prime = 1;
131
132         for (j = 2; j * j <= i; j++) {
133             if (i % j == 0) {
134                 prime = 0;
135                 break;
136             }
137         }
138
139         if (prime)
140             count++;
141     }
142
143     printf("%d", count);
144
145     return 0;
146 }
147
```

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

```
cd "/Users/apple/Documents/C_AND_CPP_MODULES/" && gcc tempCodeRunnerFile.c -o tempCodeRunnerFile && "/Users/apple/Documents/C_AND_CPP_MODULES/"tempCodeRunnerFile
apple@apples-MacBook-Pro C_AND_CPP_MODULES % cd "/Users/apple/Documents/C_AND_CPP_MODULES/" && gcc tempCodeRunnerFile.c -o tempCodeRunnerFile && "/Users/apple/Documents/C_AND_CPP_MODULES/"tempCodeRunnerFile
11 20
40
apple@apples-MacBook-Pro C_AND_CPP_MODULES %
```

Tip: Try the Plan

agent to research

plan before

implementing

changes.

week4\_ma

Describe what

EXPLORER

tempCodeRunner...  
tempCodeRunner...  
week1.cpp  
week3.cpp  
week4.cpp

week4.cpp

```
115 //question 6
116 #include <iostream>
117 using namespace std;
118
119 int main() {
120     int L, R, count = 0;
121
122     cin >> L >> R;
123
124     for (int i = L; i <= R; i++) {
125         if (i < 2)
126             continue;
127
128         bool prime = true;
129
130         for (int j = 2; j * j <= i; j++) {
131             if (i % j == 0) {
132                 prime = false;
133                 break;
134             }
135         }
136
137         if (prime)
138             count++;
139     }
140
141     cout << count;
142
143     return 0;
144 }
```

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

cd "/Users/apple/Documents/CPP\_modules/" && g++ tempCodeRunnerFile.cpp -o tempCodeRunnerFile && "/Users/apple/Documents/CPP\_modules/"tempCodeRunnerFile  
apple@apples-MacBook-Pro CPP\_modules % cd "/Users/apple/Documents/CPP\_modules/" && g++ tempCodeRunnerFile.cpp -o tempCodeRunnerFile && "/Users/apple/Documents/CPP\_modules/"tempCodeRunnerFile  
1 10  
4  
apple@apples-MacBook-Pro CPP\_modules %

### Problem Statement

Given a positive integer **N**, find the sum of all its **proper divisors**.

A **proper divisor** of a number is a positive divisor of the number excluding the number itself.

Your task is to write a program that calculates and prints the sum of all proper divisors of **N**.

### Input Format

A single integer **N**.

### Output Format

Print a single integer representing the sum of all proper divisors of **N**.

### Constraints

$$1 \leq N \leq 10^6$$

### Example 1

#### Input

12

#### Output

16

#### Explanation

The proper divisors of **12** are:

1, 2, 3, 4, 6

Their sum is:

$$1 + 2 + 3 + 4 + 6 = 16$$

### Example 2

#### Input

15

#### Output

9

#### Explanation

The proper divisors of **15** are:

1, 3, 5

Their sum is:

$$1 + 3 + 5 = 9$$

EXPLORER

C\_AND\_CPP\_MODULES

tempCodeRunner...  
tempCodeRunner...  
week2\_main.c  
week3\_main.c  
week4\_main.c

week3\_main.c

week4\_main.c X

week4\_main.c

main()

```
149 //question 7
150 #include <stdio.h>
151
152 int main() {
153     int N, i, sum = 0;
154
155     scanf("%d", &N);
156
157     for (i = 1; i < N; i++) {
158         if (N % i == 0)
159             sum += i;
160     }
161
162     printf("%d", sum);
163
164     return 0;
165 }
166
```

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

```
cd "/Users/apple/Documents/C_AND_CPP_MODULES/" && gcc tempCodeRunnerFile.c -o tempCodeRunnerFile && "/Users/apple/Documents/C_AND_CPP_MODULES/"tempCodeRunnerFile
apple@apples-MacBook-Pro C_AND_CPP_MODULES % cd "/Users/apple/Documents/C_AND_CPP_MODULES/" && gcc tempCodeRunnerFile.c -o tempCodeRunnerFile && "/Users/apple/Documents/C_AND_CPP_MODULES/"tempCodeRunnerFile
20
22
apple@apples-MacBook-Pro C_AND_CPP_MODULES %
```

Code

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Tip: Try the Plan agent to research plan before implementing changes.

week4\_main

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EXPLORER

week1.cppweek3.cppweek4.cpp X

CPP\_MODULES

tempCodeRunner...  
tempCodeRunner...  
week1.cpp  
week3.cpp  
week4.cpp

week4.cpp > ...

147  
148  
149 //question 7  
150 #include <iostream>  
151 using namespace std;  
152  
153 int main() {  
154 int N, sum = 0;  
155  
156 cin >> N;  
157  
158 for (int i = 1; i < N; i++) {  
159 if (N % i == 0)  
160 sum += i;  
161 }  
162  
163 cout << sum;  
164  
165 return 0;  
166 }  
167

PROBLEMSOUTPUTDEBUG CONSOLETERMINALPORTS

Code + - - - - X

cd "/Users/apple/Documents/CPP\_modules/" && g++ tempCodeRunnerFile.cpp -o tempCodeRunnerFile && "/Users/apple/Documents/CPP\_modules/"tempCodeRunnerFile  
apple@apples-MacBook-Pro CPP\_modules % cd "/Users/apple/Documents/CPP\_modules/" && g++ tempCodeRunnerFile.cpp -o tempCodeRunnerFile && "/Users/apple/Documents/CPP\_modules/"tempCodeRunnerFile  
12  
16  
apple@apples-MacBook-Pro CPP\_modules %

Tip: Try the Plan agent to research plan before implementing changes.

week4.cpp

Describe what

### Problem Statement

Given a positive integer **N**, print all the factors of the number in **ascending order**.

A factor of a number is an integer that divides the number completely without leaving any remainder.

Your task is to write a program that prints all the factors of **N**.

### Input Format

A single integer **N**.

### Output Format

Print all the factors of **N** in ascending order, separated by spaces.

### Constraints

$$1 \leq N \leq 10^6$$

### Example 1

#### Input

12

#### Output

1 2 3 4 6 12

#### Explanation

The numbers **1, 2, 3, 4, 6**, and **12** divide **12** exactly without leaving any remainder.

### Example 2

#### Input

25

#### Output

1 5 25

#### Explanation

The numbers **1, 5**, and **25** are the only factors of **25**.

FILES

CodeRunner...

CodeRunner...

k1.cpp

k3.cpp

k4.cpp

C: week4.cpp &gt; main()

```
168
169
170
171 //question 8
172 #include <iostream>
173 using namespace std;
174
175 int main() {
176     int N;
177
178     cin >> N;
179
180     for (int i = 1; i <= N; i++) {
181         if (N % i == 0)
182             cout << i << " ";
183     }
184
185     return 0;
186 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
cd "/Users/apple/Documents/CPP_modules/" && g++ tempCodeRunnerFile.cpp -o tempCodeRunnerFile && "/Users/apple/Documents/CPP_modules/"tempCodeRunnerFile
apple@apples-MacBook-Pro CPP_modules % cd "/Users/apple/Documents/CPP_modules/" && g++ tempCodeRunnerFile.cpp -o tempCodeRunnerFile && "/Users/apple/Documents/CPP_modules/"tempCodeRunnerFile
18
1 2 3 6 9 18
apple@apples-MacBook-Pro CPP_modules %
```

